Data Science →

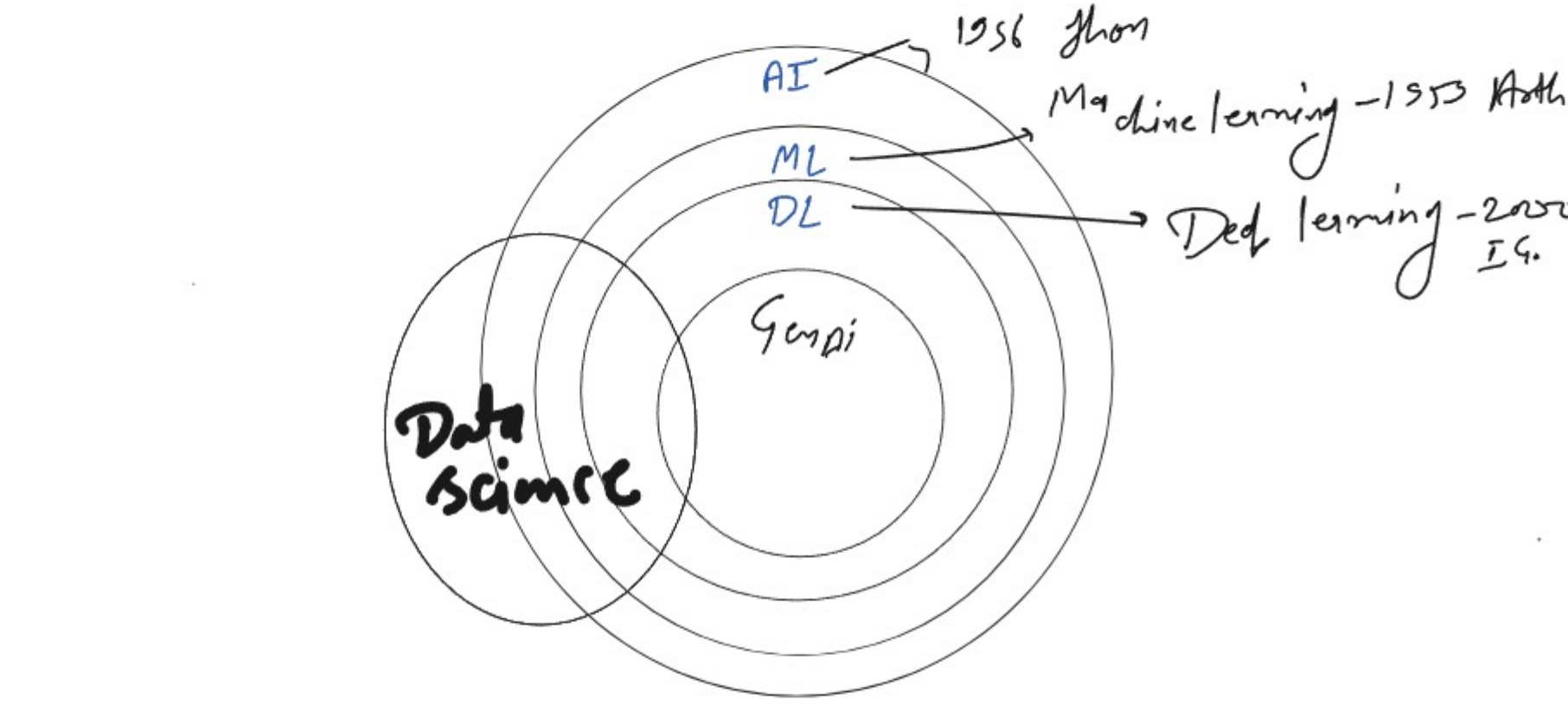
Data

Science

data

Study

Study of data that is called data science



Machine learning

Machine

learning

(i) Sup.

(ii)

(iii) Rein

Machine learning

Machine

learning

types

(i) Supervised => labels

(ii) Unsupervised

(iii) Reinforcement learning

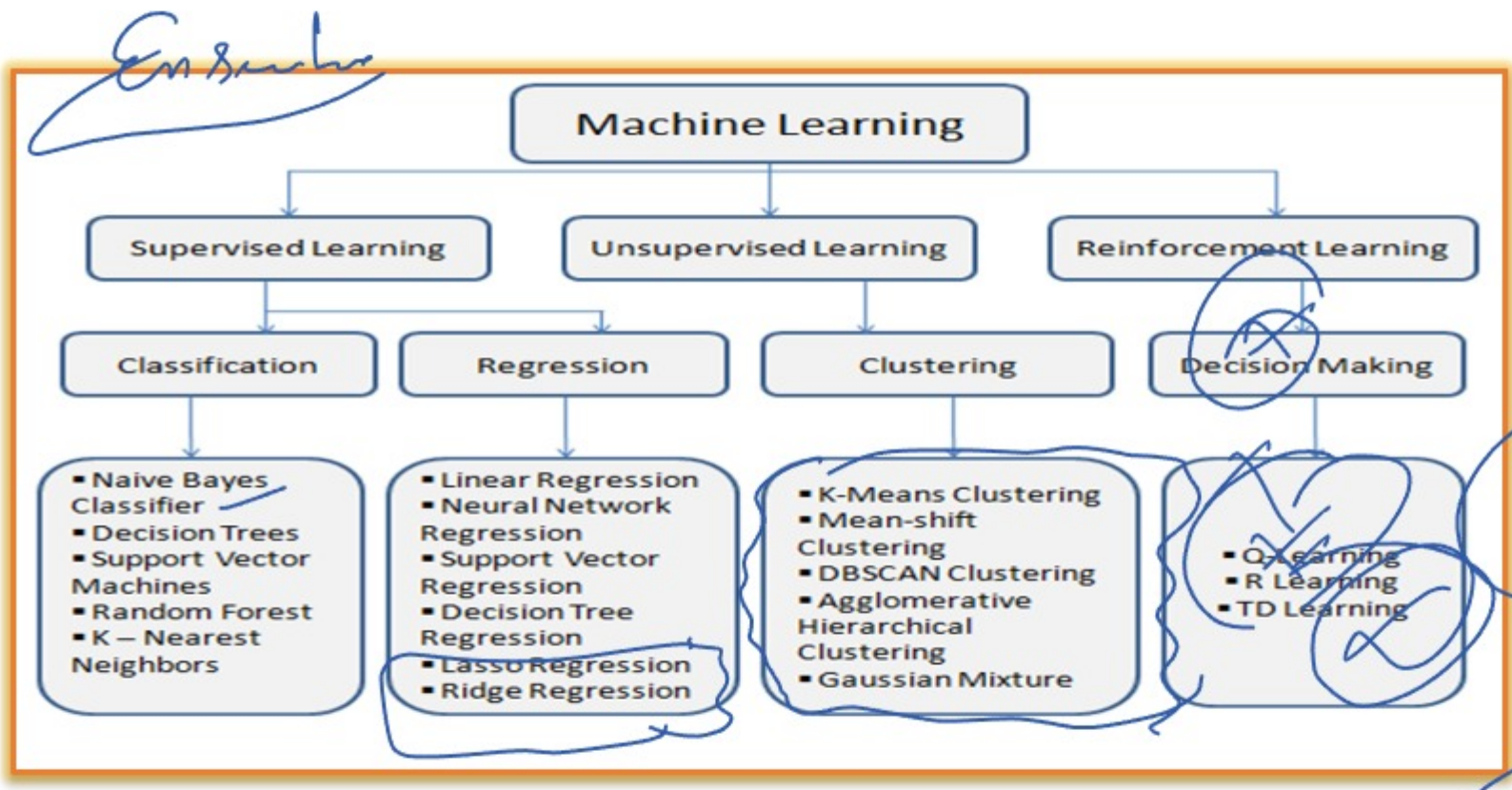
(iv) Semi supervised

(1)

data

data Collect

learning → Reference, EDA,



Classification

Positive / neg

True / False

Yes / No

Regression

Ensemble

Ada

Row

xg

xg

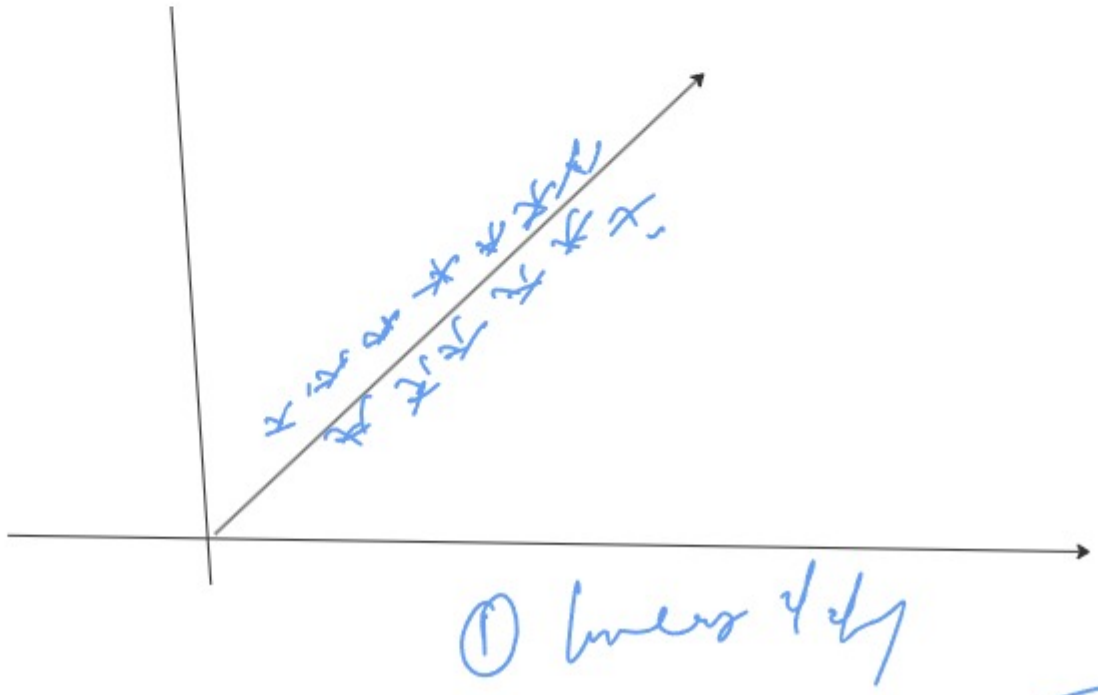
Bagging

Boosting

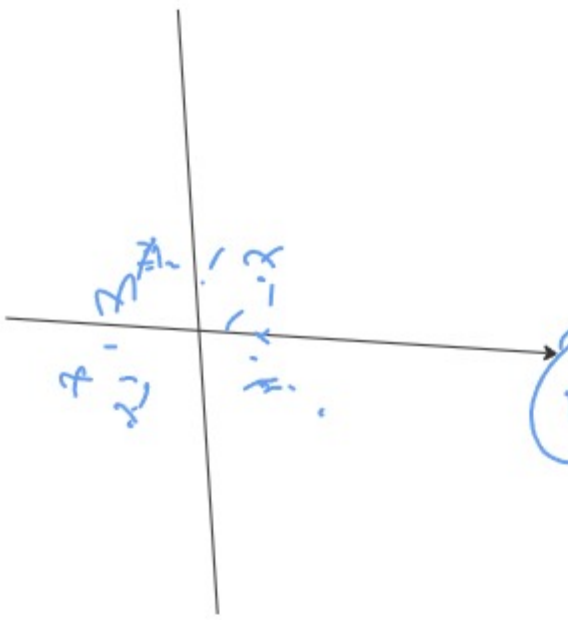
Stacking

Linear Regression
supervised
Regression
labels

Linear and non linear:



① linear



② non linear

ML-DL

Linear Regression

$y = mx + c$

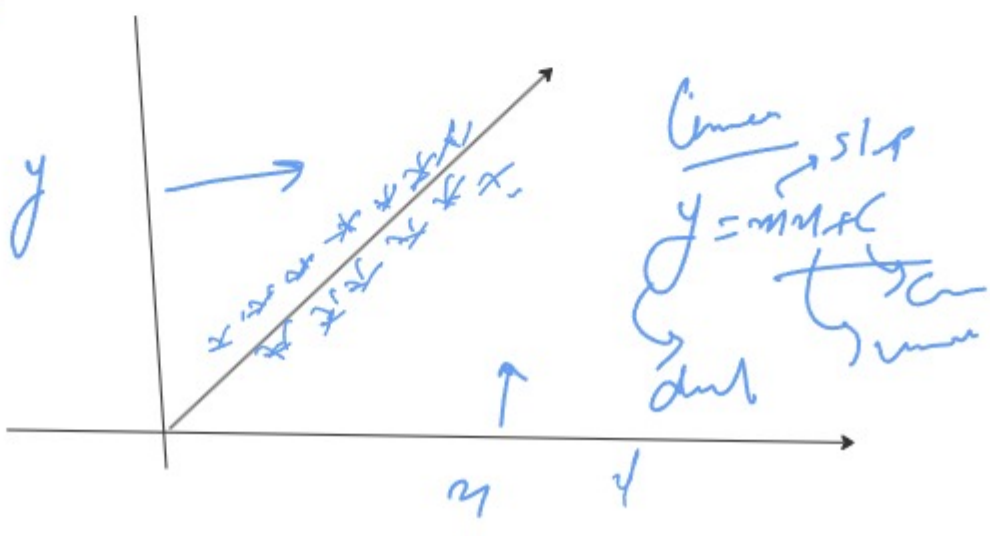
slope, constant

independent variable

dependent variable

Bike Price

Brand_name, age, kms, owner, Pow



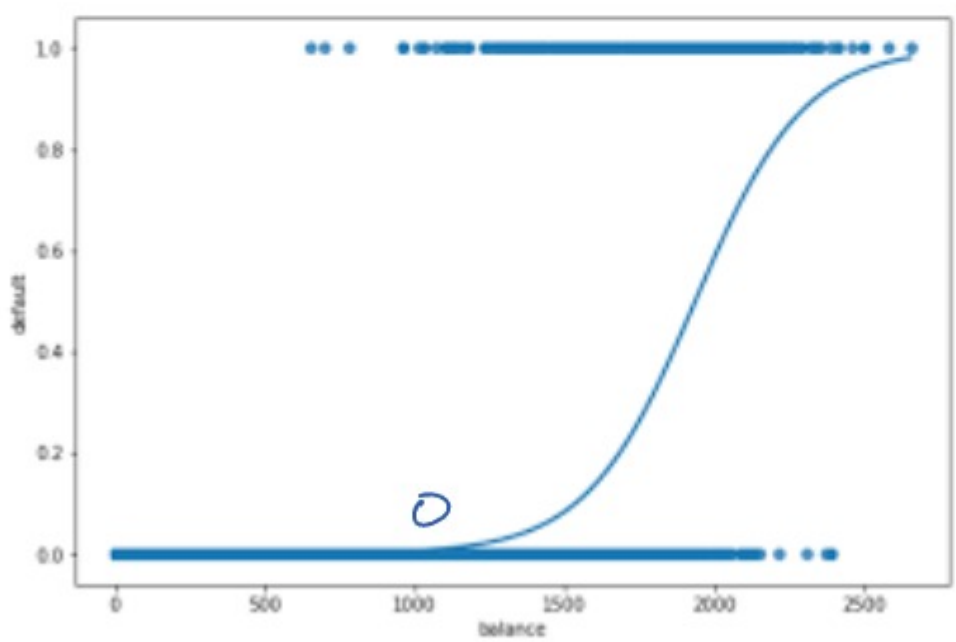
Logistic Regression / AWN /

Sup. , # labled

Yes / No (1/0

↓

Sigmoid



1 Sigmoid

0 / 1

output

true / False

$$\text{formula} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

output.

e = 2.71...

SVM

(Support vector machine)

Supervised

label

classification (mainly)

&

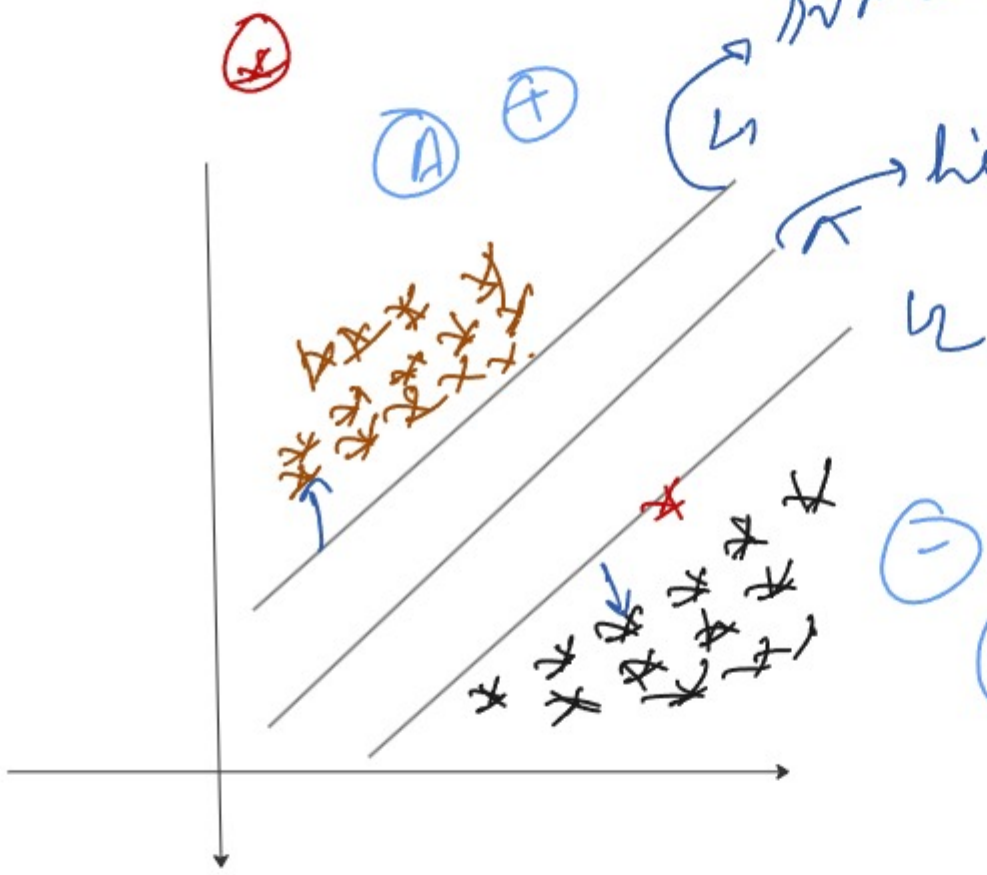
Regression

SVMR & SVMC

types

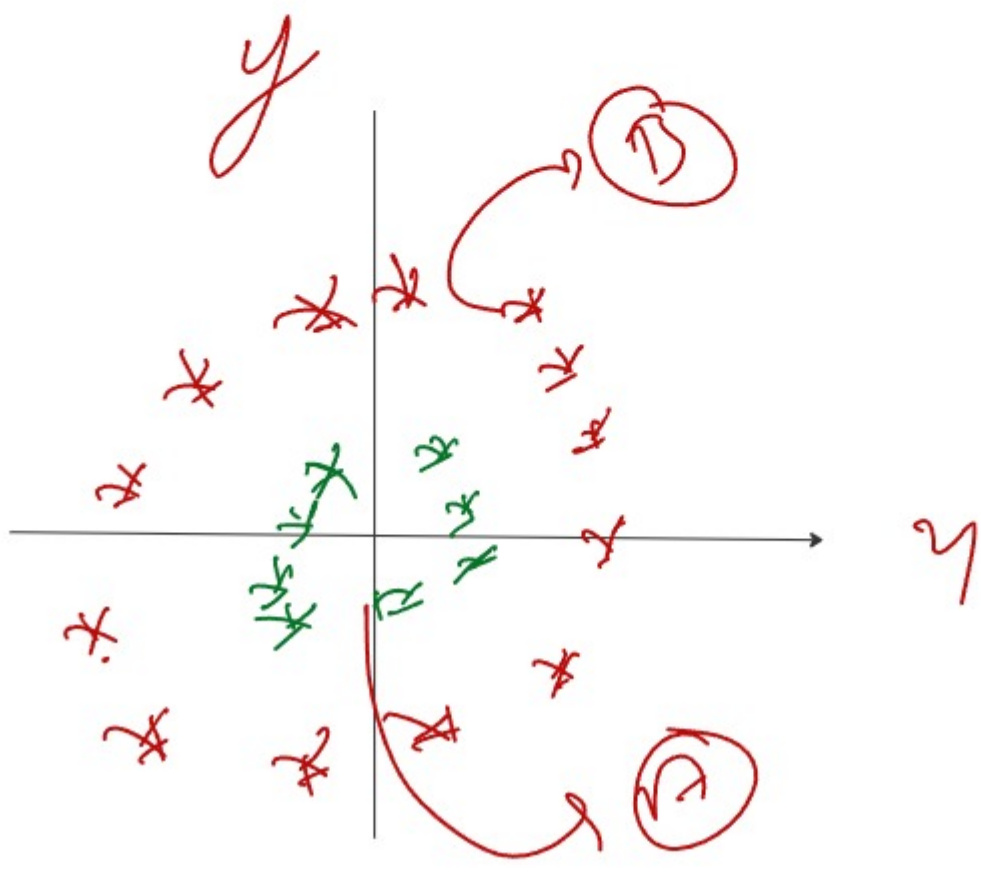
① Linear

② Non linear



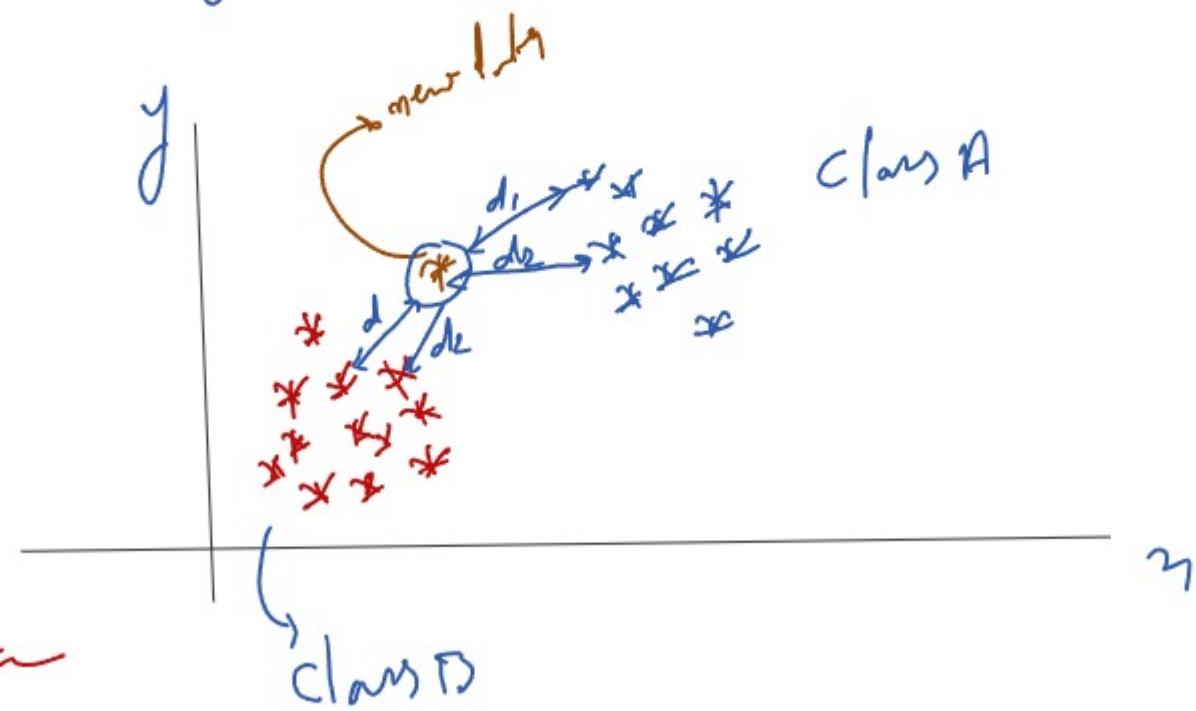
$y = mx + c$

$(w^T + y)$



KNN (K-Nearest Neighbor)

* Sup.
* K = ?
* classification (mainly)
  &
  Regression



Euclidean distance

$$(x,y) = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$$

2. Manhattan distance

$$d(x,y) = \sum_{i=1}^{n} |x_i - y_i|$$

3. Minkowski distance

$$d(x,y) = \left(\sum_{i=1}^{n} (x_i - y_i)^p \right)^{1/p}$$

$p = 2$

$p = 1$

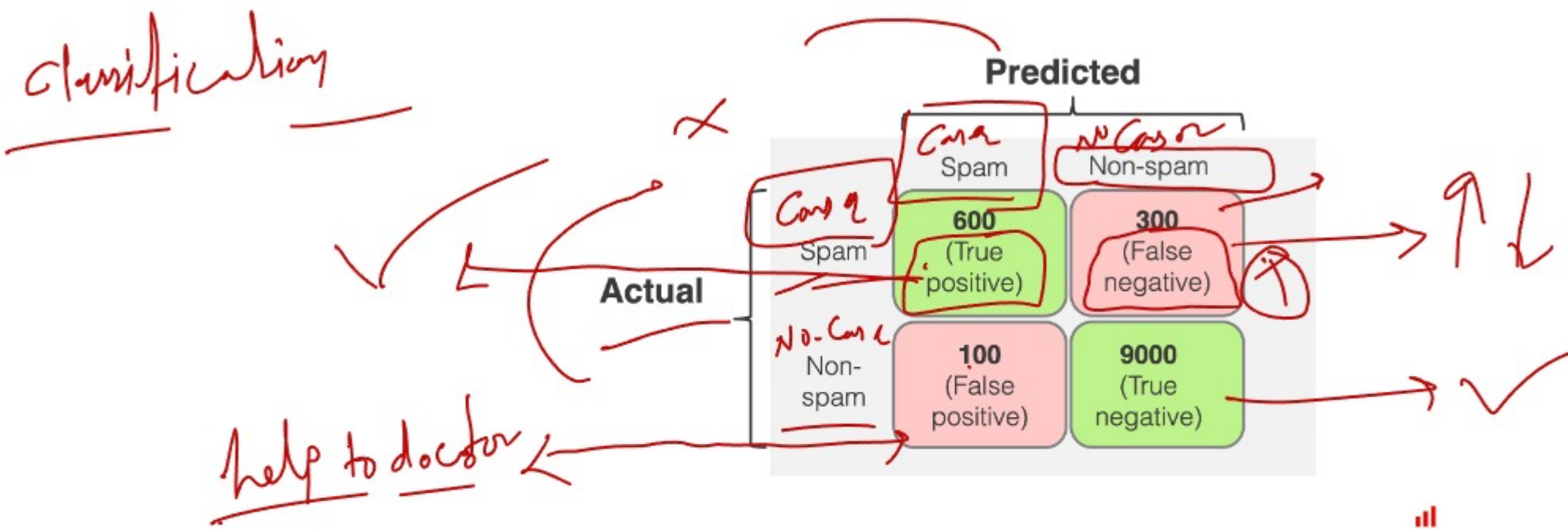
Recommendation
Spam detection

Advantage

Simple to use
no training step
Parameter

dis slow (↑) dataset (↑)
overfitting

classification report



Performance Metrics /
Model Evaluation Techniques
classification

Accuracy
Precision
Recall
f1 score
Roc - Auc
Regression metirics
MAE
MSE
RMSE
HUBBER loss

Accuracy

$$= \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

$$= \frac{TP}{TP + FP}$$

Recall

$$= \frac{TP}{TP + FN}$$

F1 Score

$$\hookrightarrow = 2 \times \frac{Pre \times Recall}{Pre + Recall}$$

Regression metric

(i) MSE
(ii) MAE
(iii) RMSE
(iv) Huber loss

(i) MSE (mean squared Error)

$$MSE = \frac{1}{n}\sum (y_i - \hat{y}_i)^2$$

(ii) MAE (Mean Absolute error)

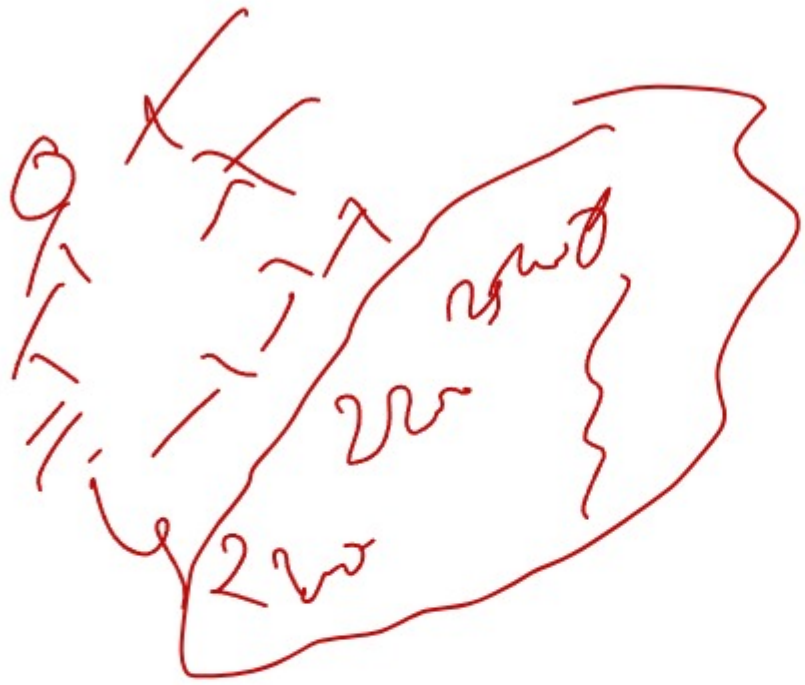
$$MAE = \frac{1}{n}\sum |y_i - \hat{y}_i|$$

RMSE = (Root mean squared error)

$$RMSE = \sqrt{MSE}$$

Huber loss = MSE + MAE

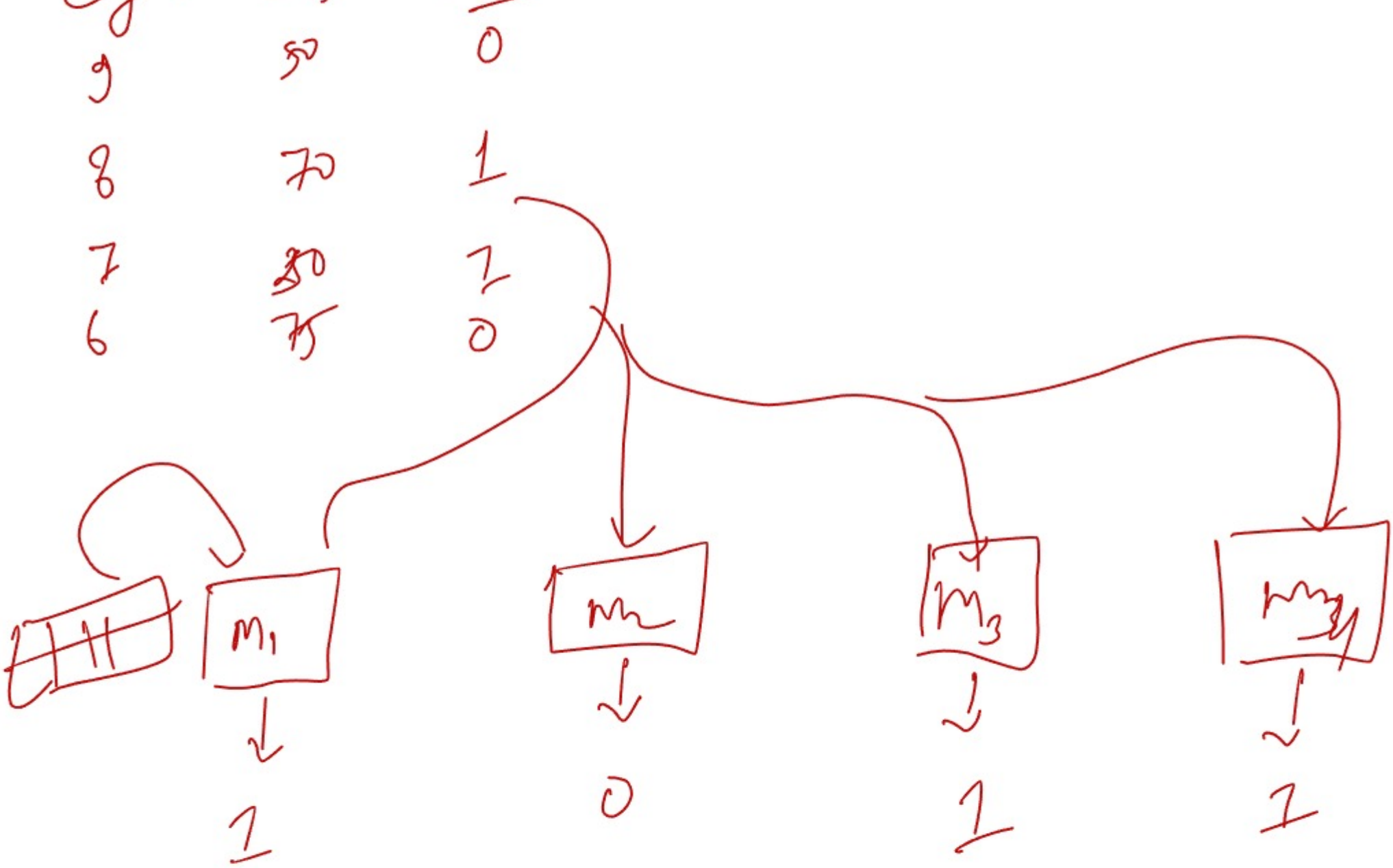
Ensemble learning



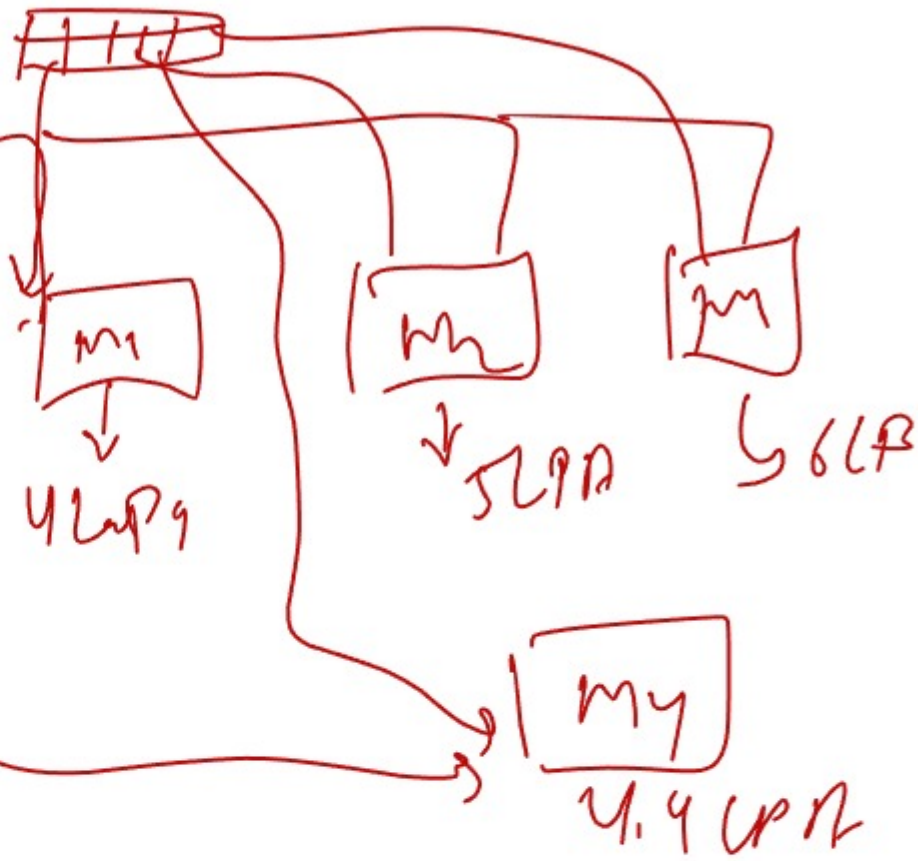
2400 kg

2350 kg

CGPA	IQ	Placement
9	50	0
8	70	1
7	50	1
6	75	0



cgpa	IQ	LPA
6	50	5 LPA
7	100	6 LPA
8	70	6.5 LPA
♡	100	7 LPA



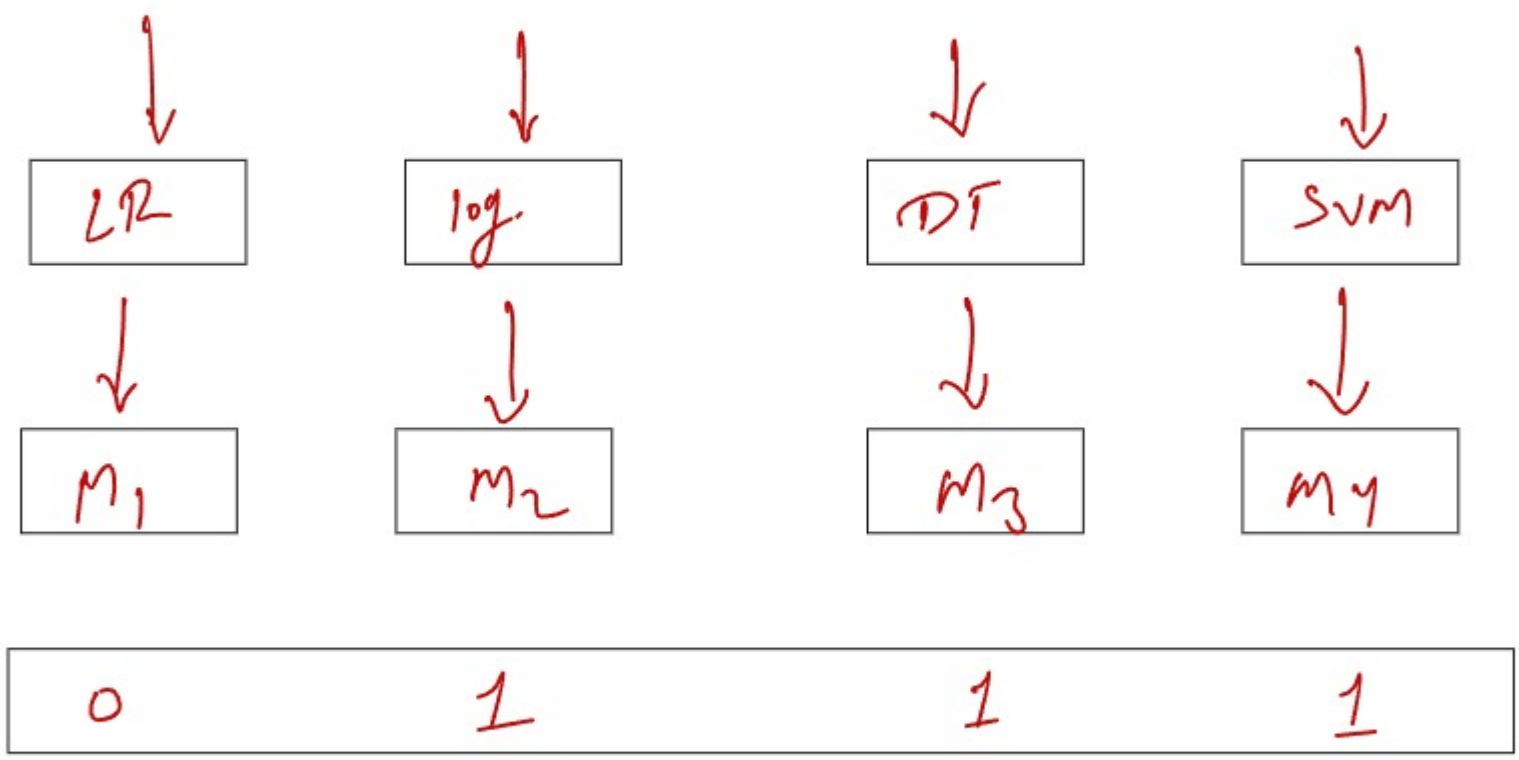
$$\frac{4+5+6+4.5}{4} =$$

① Bagging → Algo Same ⇒ Same model
② Boosting
③ voting → Algo diff ⇒ model diff.
④ Stacking

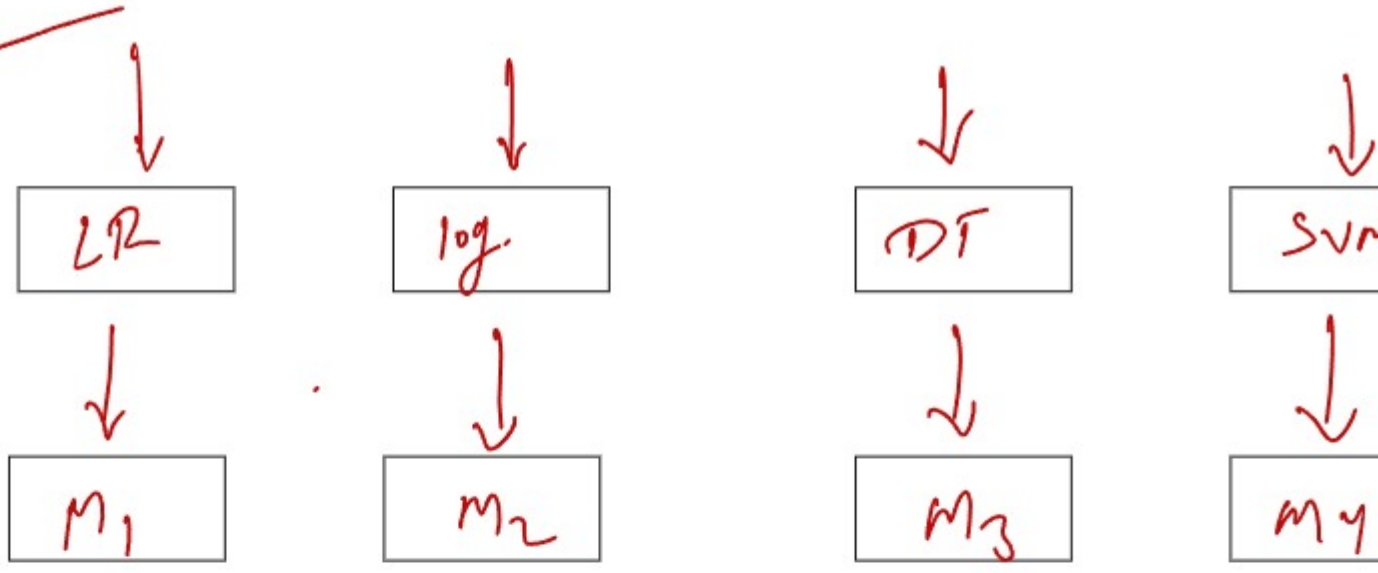
formula ensemble
for voting

Voting Algo diff.

from sklearn.ensemble import votting classifer , stacking



Stacking



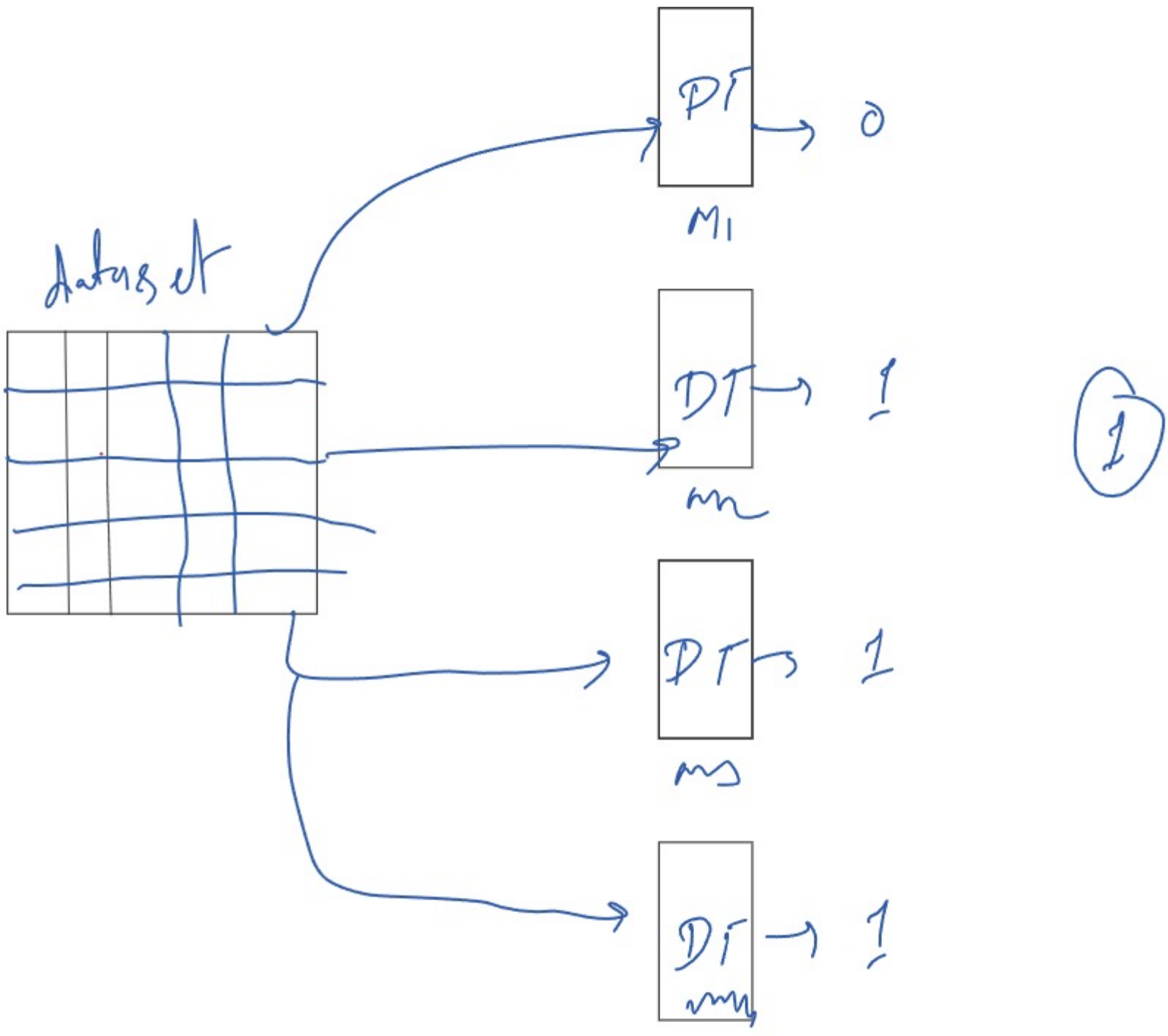
Alg.	Pred
LR	0
log	1
DT	1
SVM	1

Bagging & Boosting

Algo - Same, Model - Same

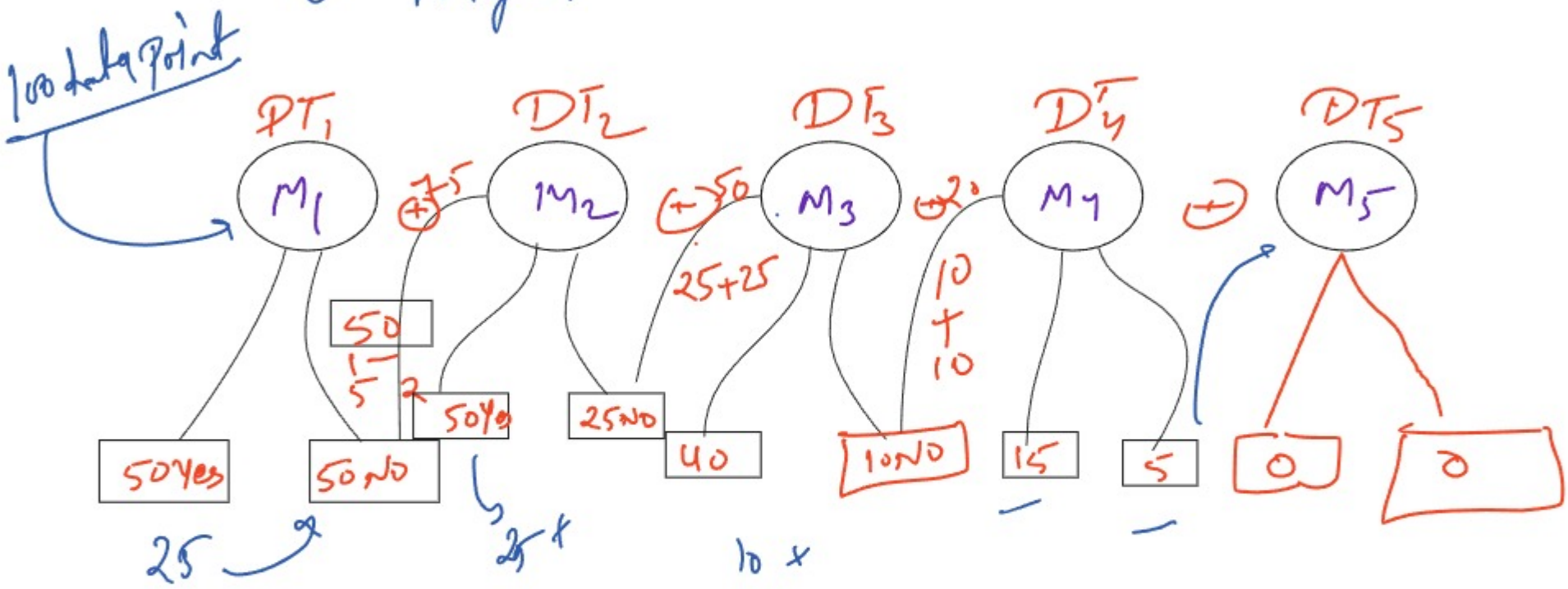
Bagging Ensemble learning

Algo Same



(11) Random forest

Boosting → Algo.



Algo

Adaboost

xGboost

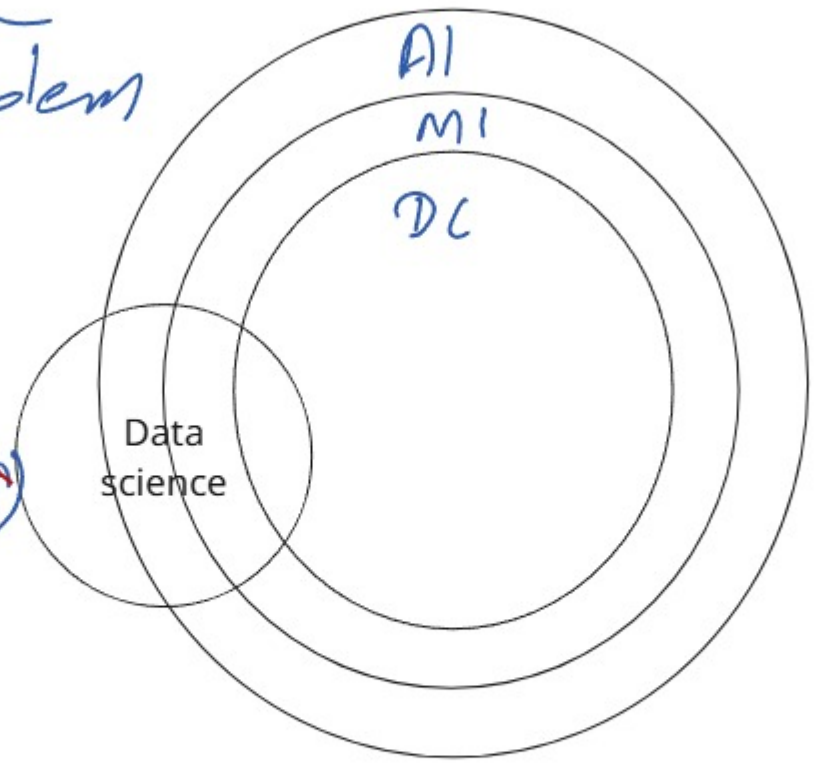
Gradient boosting Algo

Deeplearning

Subset of ML
Complex problem solve

dataset (↑)

Accuracy (↑)



ML

dataset (↑)

Accuracy (↓

score (↓)

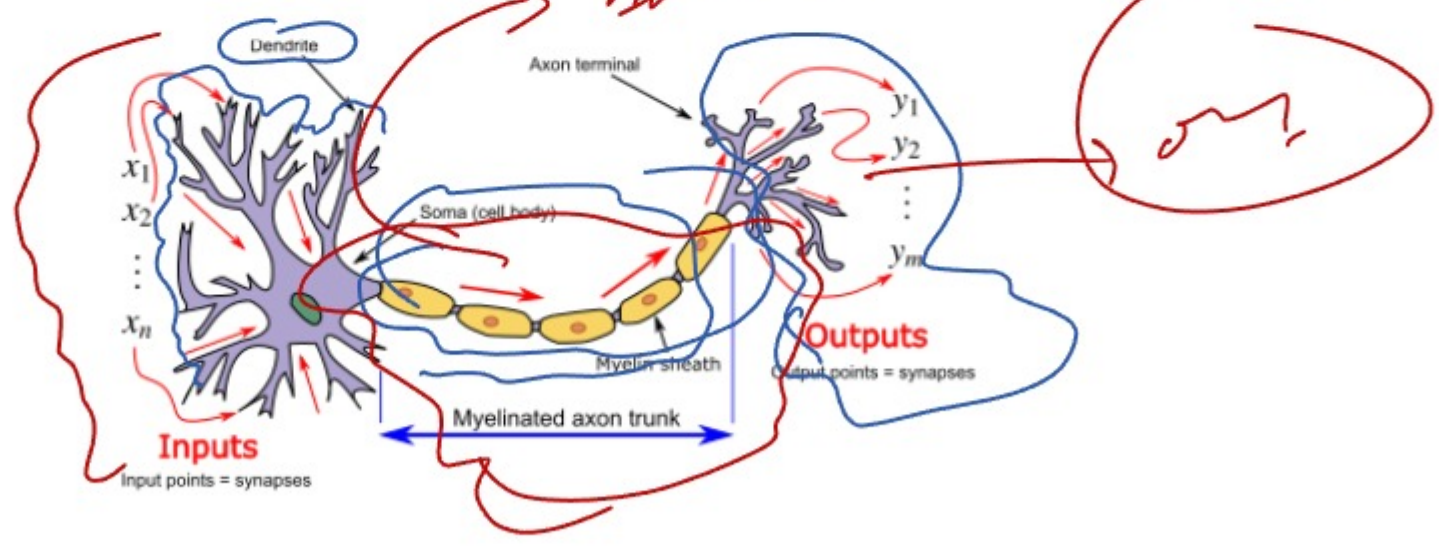
Algo of DL

(i) ANN (Artificial Neural network)
(ii) CNN (Convolutional neural network)
(iii) RNN (Rec " ")

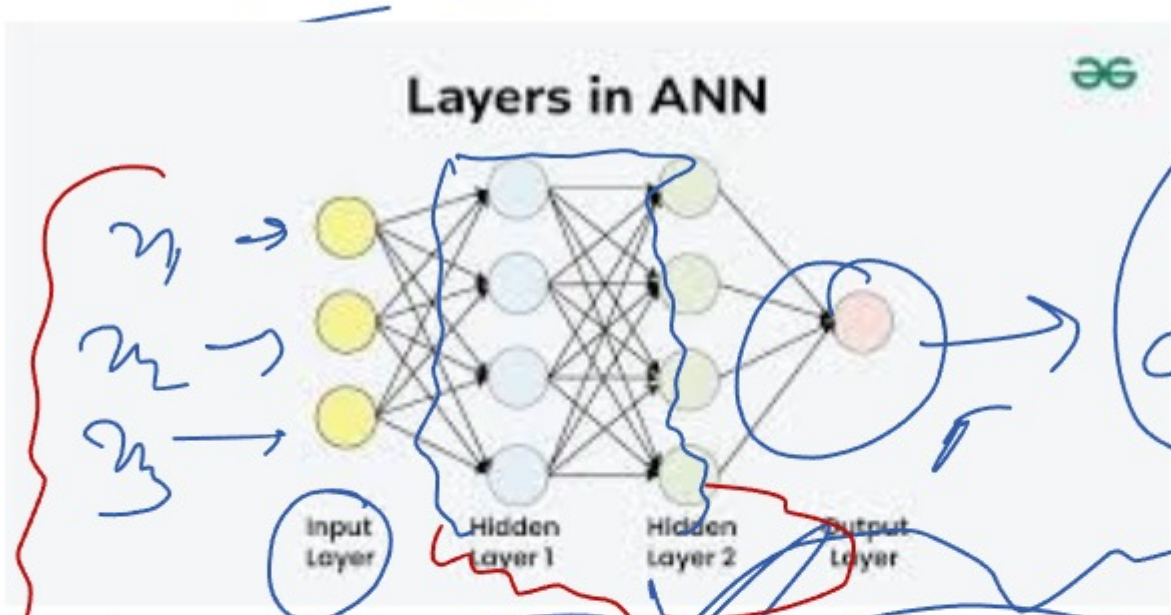
LSTM
GRU

ANN Artificial Neural Network

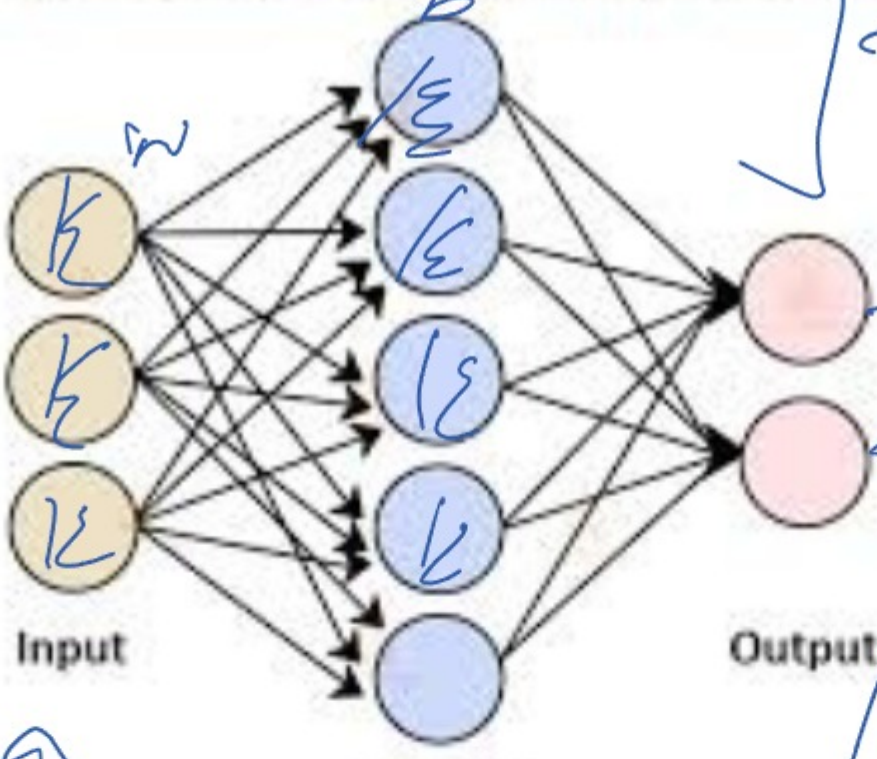
Biological Neuron



ANN



Architecture of Artificial Neural Network



CNN

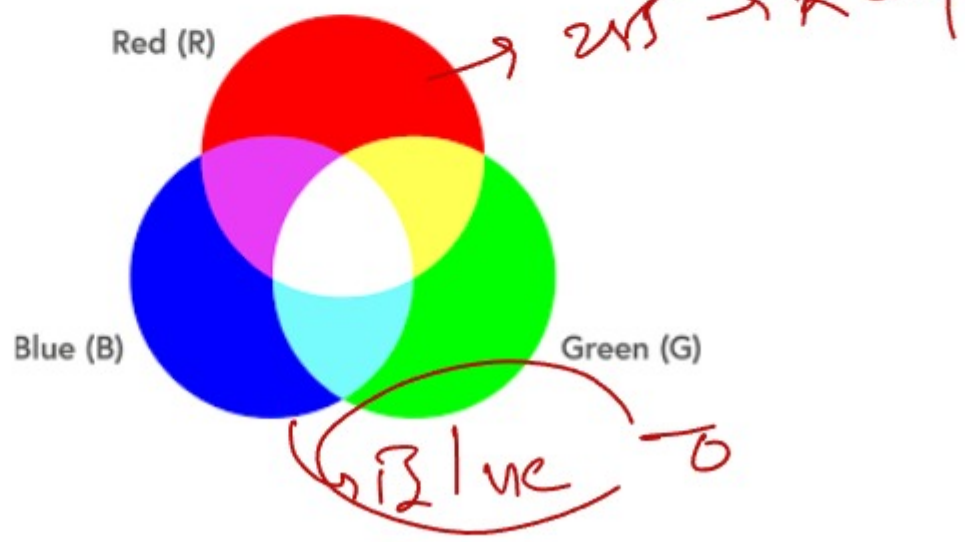
Image

Image
Vedio
Animation

group of Pixel

RGB
R → Red
G → Green
B → Blue

RGB (0 to 255)
Black/White (0, 1)



255 → Red

Blue → 0

Black

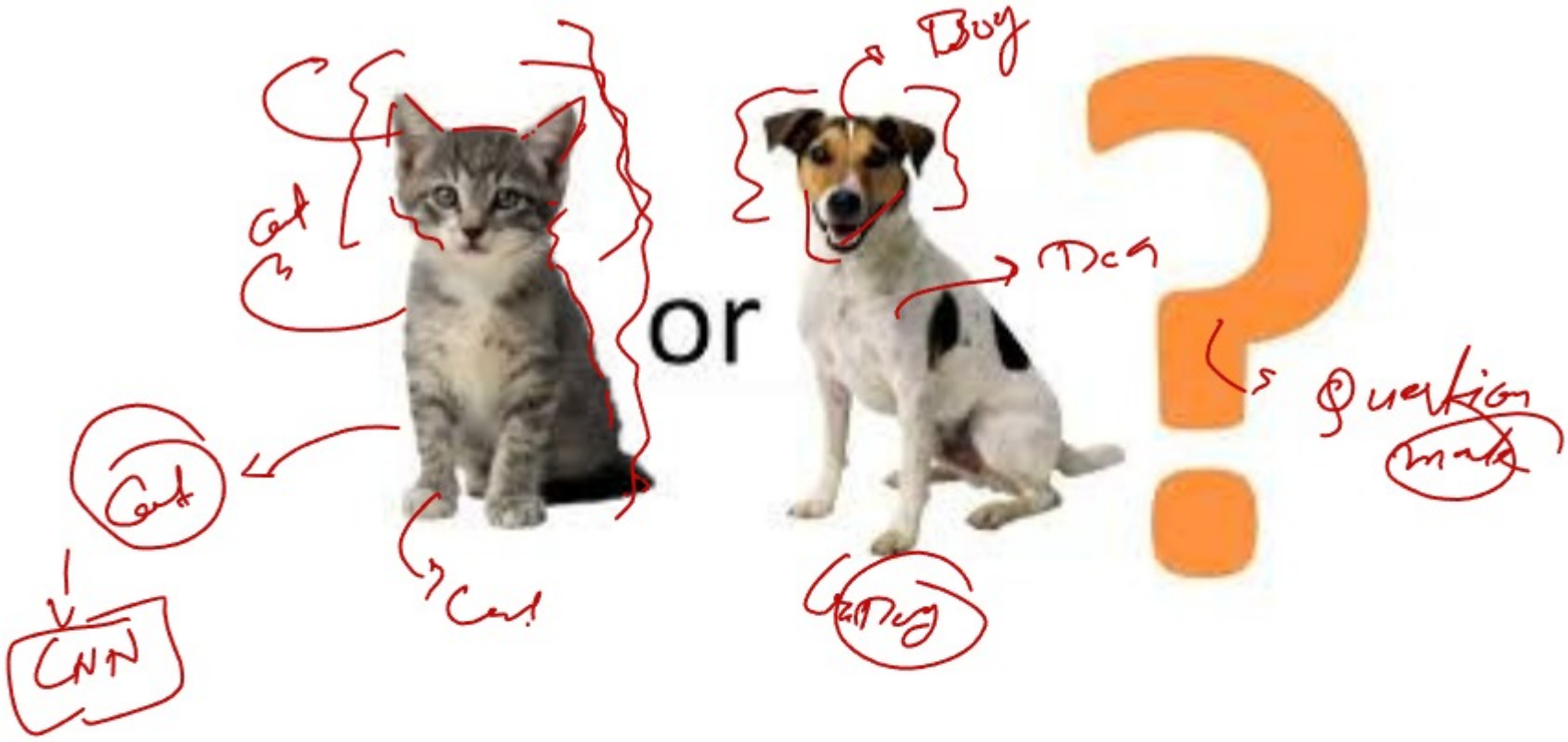
-1

→ +1 White

3×3

+1	+1	-1
+1	-1	+1
-1	+1	+1

⇒

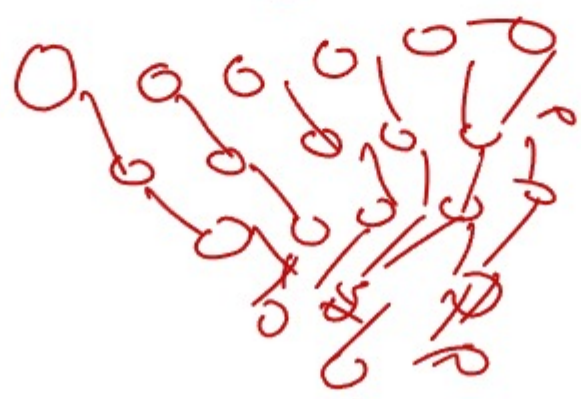


CNN Layer

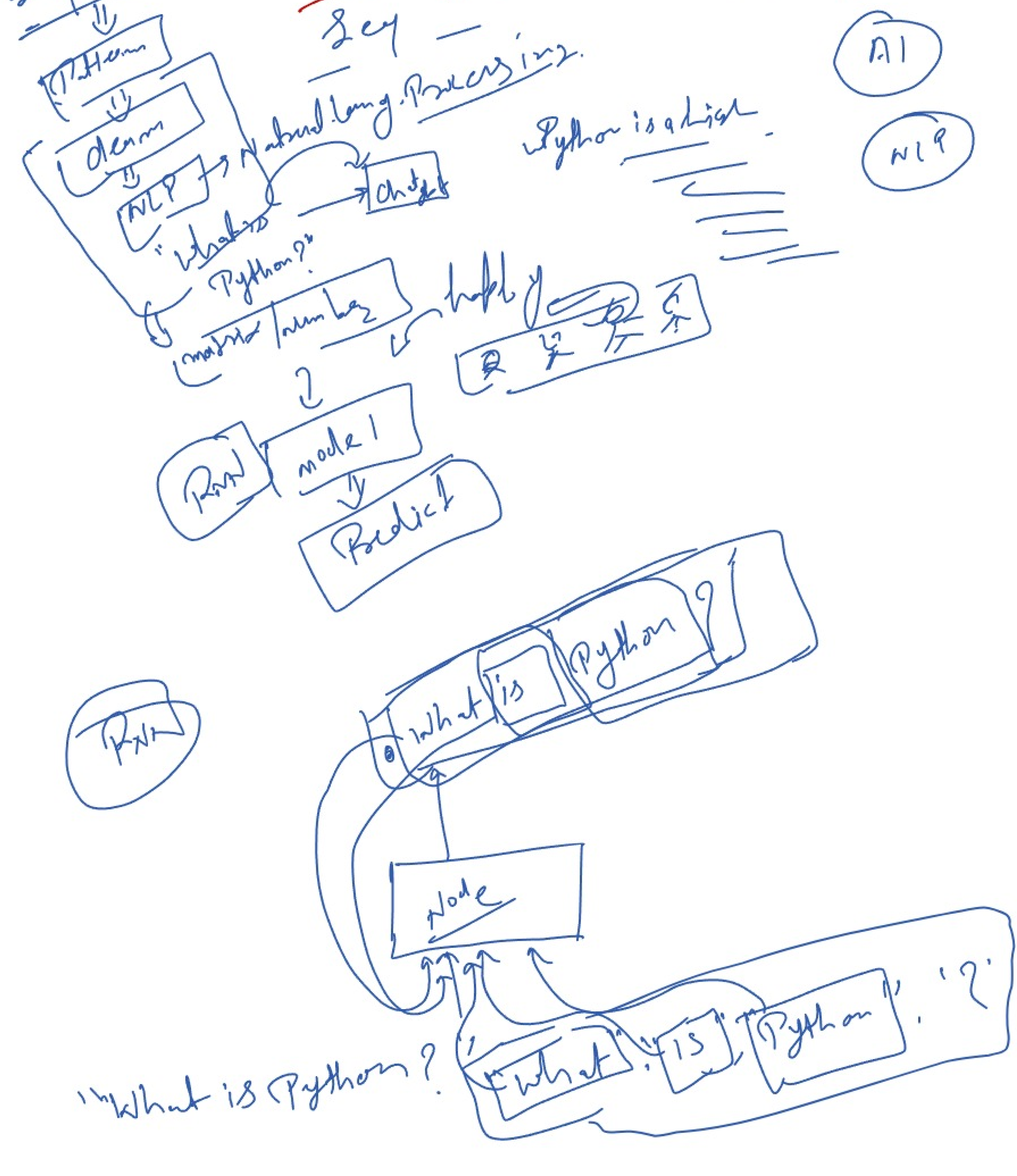
1. Convonal layer
2. Pooling layer
3. Flatten layer

Pixel

Numpy ⇒ Flatten()
Ravel()



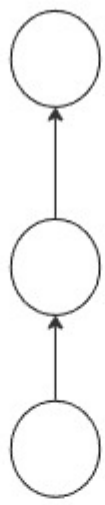
RNN - (Recurrent Neural network)



① Types of RNN

(i) one to one RNN
(ii) one to many
(iii) many to one
(iv) many to many
(v) Lstm
(vi) GRU

(I) one to one RNN

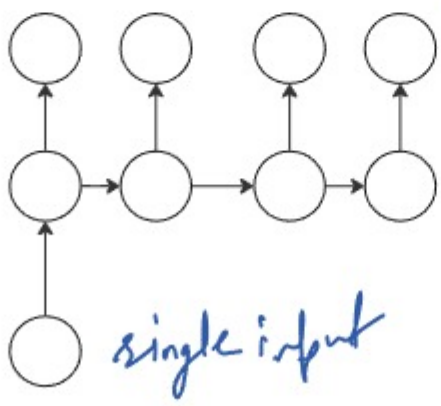


single output

single input

(II) one to many RNN

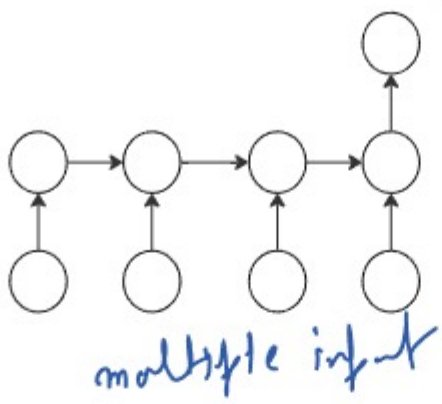
multiple output



single input

(III) Many to one RNN

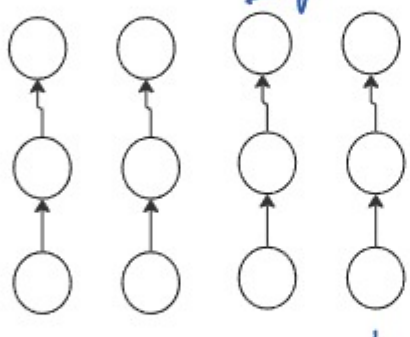
one output / single



multiple input

(IV) many to many RNN

many output



many i/p

LSTM

RNN

[Long Short term memory]

[Vanishing Gradient Problem]

Advanced of RNN

Application of Lstm

(I) lang. trans

(II) Speech Rec

Lstm Architecture

(I) Input gate → Add

(II) forget gate → Remove

(III) output gate → output provide

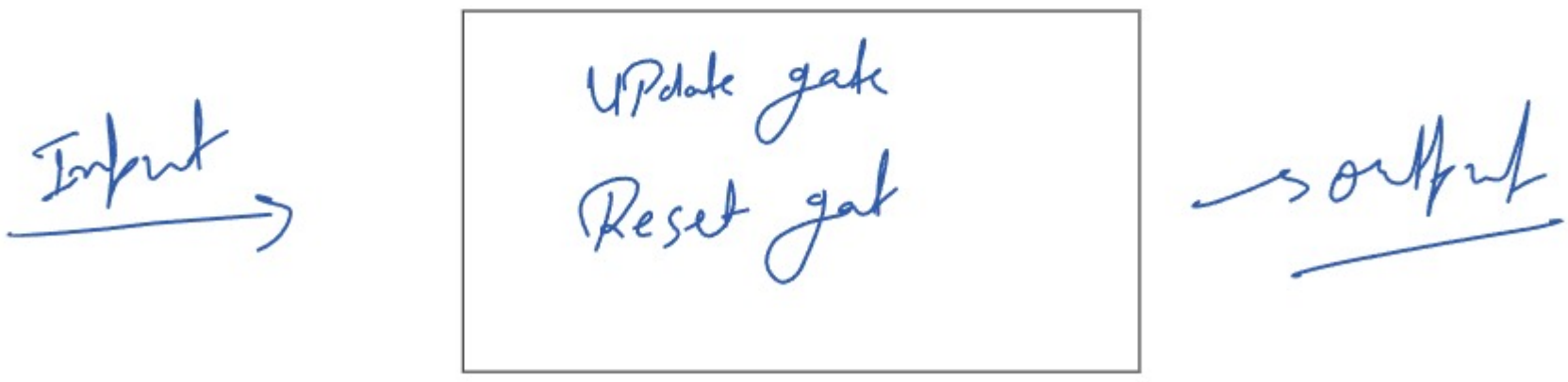
GRU (Gated Recurrent Units)

faster and Stable than lstm
Train time(-ve)
Gradient Problem

GRU Arti

(I) Update gate

(II) Reset gate



NLP
(Natural lang. Processing)

Subfield of ai

Text classification

Tags

NLP →

Text preprocessing steps

① Lower casing
② HTML removal
URL
@ # $
Emoji

Text vectorization

Text conversion into numbers

(I) OHE
(II) BOW (Bag of word)
(III) N-Grams
(IV) TFIDF
(V) Custom feature

Bag of Words

Agr two class are same to yr number same
assign

* Order are not matter
* Context do not matter